The Metric Is Not the Construct

Construct Validity at Every Layer of a Lending Audit

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ABSTRACT

The same model, evaluated on the same applicants, can pass or fail a fairness audit depending on settings that appear nowhere in the audit's output. This paper identifies those settings across a complete audit of consumer lending (1,000 credit applications and 10,978 Georgia mortgage applications) and measures how often each one changes the verdict. Each is an instance of one underlying event: a construct someone intended ("favorable outcome," "the protected class," "refusal") is operationalised as something a program can compute, the two differ, and the output looks equally clean either way. Nearly every wrong verdict we measured erred in the same direction: toward the clean, reassuring reading.

The settings live at three layers. In the metric library (verified in AIF360 0.6 and Fairlearn 0.13): one AIF360 constructor defaults `favorable_label=1.0`, so on a target named `loan_default` it reports the disadvantaged group as favoured, with DI = 1.2072 where the oriented value is 0.9862; and neither library attaches an interval, a sample size, or a record of the reference group to a disaggregated ratio. In the mapping from attribute to construct: a banded age attribute reads 0.9791 where the 62+ cut (the boundary the mortgage data ship a dedicated flag for) reads 0.8897, the largest age disparity present and a different question, answered equally cleanly; and a subsampling study over the full mortgage population shows that at the audit size we originally used, a no-floor audit falsely clears a subgroup that sits below the screen at full population in 83.8% of draws while a reporting floor never does, because it never reports the subgroup at all and misidentifies the worst-treated group in 100% of draws. In the audit's own instruments: a refusal detector that operationalises "refusal" as *under 40 characters* flagged six correct answers; a score mean taken mostly over empty API results recorded a provider as permanently blocked; and a cross-model "prompt sensitivity" spread is concentrated in the terse-output cycle, where our own scorer penalises instructed brevity.

The implication is that a fairness verdict is not a property of the model alone; it is a property of the model plus an unrecorded configuration. The same data yield "worst-treated group at 0.67," "at 0.90," or "unknowable," depending on choices no reader can see. An audit should therefore ship its configuration the way results ship error bars. We close with the six-item record that would accomplish this; every number in the paper maps to a released artifact that regenerates it.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Social and professional topics** → *Governmental regulations*; • **Human-centered computing** → *Human computer interaction (HCI).*

KEYWORDS

measurement, construct validity, algorithmic auditing, disparate impact, intersectionality, lending, large language models

1 INTRODUCTION

A fairness verdict is not a property of a model alone. It is a property of the model plus a configuration: which outcome counts as favorable, who counts as protected, whom a disparity is measured against, how much evidence the number rests on, and which instruments check the audit's own components. None of those settings appear in the audit's output, and each is an instance of one underlying event: the construct a fairness instrument is *named for* differs from the construct it actually *operationalises* [23], the output is clean either way, and nothing records the substitution. This paper measures how often each such setting changes the verdict; nearly every wrong verdict we measured erred in the same direction, toward the clean, reassuring reading.

Lending is the right place to make that measurement. Two credit decisions dominate household outcomes: consumer installment credit, which governs access to short-term liquidity, and residential mortgage origination, which governs access to the principal asset most households will ever own. Lee et al. [29] make the organising commitment of human-centered fair AI that fairness is defined relative to a human decision context rather than in metric space. Lending's decision context is unusually well documented: the mortgage data are published for public audit, the protected categories come with written definitions, and the datasets themselves ship the flags those definitions turn on. That is what turns the gap between named and operationalised construct from something arguable into something measurable.

The gap appears at every layer of an audit, takes the same shape at each layer, and is invisible in the output. The layers are: the metric library, whose defaults resolve normative questions no caller answered; the mapping from attribute to construct, where the quantity computed and the quantity the audit's readers care about part company; and the audit's own evaluation instruments (detectors, gates, scores), each of which operationalises a judgement as a mechanical proxy. Table 1 collects every instance.

That the instrument is part of the finding is by now well documented. Deng et al. [13] find practitioners using fairness toolkits in ways their designers did not anticipate; Lee and Singh [30] map systematic gaps in what open-source toolkits support; Holstein et al. [22] report that practitioner needs diverge sharply from what the literature supplies. Mei et al. [35] show, in a speech setting, that audit *pipelines* introduce pitfalls that change conclusions, and Zaccour et al. [49] show that quantitative audits are more often blocked by data access than by method. We add a setting where the named constructs come with written definitions and published flags.

Contributions. (i) Three library-layer gaps verified in AIF360 0.6 and Fairlearn 0.13: a favorable-label constructor default that inverts a ratio's meaning, the absence of any uncertainty quantification on disaggregated ratios, and the absence of any record of the reference group (§4.1, appendix B). (ii) Two definition-layer gaps measured on real lending data: the banded-age operationalisation cannot represent the 62+ boundary the data's own flag encodes, and outcome-rate tooling is structurally silent about the model's inputs (§4.2). (iii) A subsampling study (2,000 draws at each of five audit sizes, five reporting policies, three reference conventions) that converts this audit's founding anecdote into a rate: policies that report small cells falsely clear violating subgroups in 84–100% of draws at the audit size we originally used, and policies that suppress them never name the worst-treated group at all (§4.3.2, appendix C). (iv) Five instrument-layer gaps in our own released harness, reported as evidence about building an audit, including a reanalysis showing that a cross-model prompt-sensitivity difference we previously reported is partly an artifact of how our scorer treats terse output (§4.4).

2 RELATED WORK

Measurement. Jacobs and Wallach [23] supply the vocabulary this paper instantiates: fairness quantities are *measurement models* of unobservable theoretical constructs, and the question to ask of any number is whether the operationalisation matches the construct it claims to measure, which is construct validity. Passi and Barocas [39] locate the same gap upstream, in problem formulation; Selbst et al. [45] describe the failure of porting formal criteria across social contexts; Obermeyer et al. [38] give the canonical in-vivo case, a proxy (cost) silently standing in for a construct (need). Disaggregated evaluation has its own choices (which groups, which cells, which comparisons) that Barocas et al. [1] catalogue; Chen et al. [8] supply the bias–variance decomposition that governs what a subgroup estimate can support, the statistical half of our §4.3. On intersectionality, Crenshaw [12] names the construct our `race×sex` cells operationalise; Buolamwini and Gebru [6] give the canonical empirical audit, Foulds et al. [17] a formal definition, and Kearns et al. [25] the auditing-theoretic treatment. That fairness criteria conflict is settled [9, 20, 27]; Corbett-Davies and Goel [11] catalogue how measures mislead when detached from decision context.

The documented decision context. We make no legal claims in this paper: discrimination law does not reduce to a metric threshold [2, 47], and adjudicating it is not our competence. What lending supplies, and what this paper uses, is narrower: the decision context comes with *written definitions and published flags.* The public mortgage data ship a dedicated `applicant_age_above_62` field because 62+ is a defined boundary in that domain's own documentation, so whether an age operationalisation can represent that boundary is a checkable question of measurement, not of law. Likewise, whether a model consumes a protected attribute directly is a documented property of its inputs (Gillis [19] examines input scrutiny at length), and no outcome-rate metric reports it (§4.2.3).

The audit's instruments. The third layer is newer but not new. LLM-as-judge evaluation is known to be fragile [51]; Wester et al. [48] study how models deny requests, the construct our refusal detector was built to measure; Sclar et al. [44] show scores move under semantically null formatting changes, the construct our prompt-sensitivity comparison was built to measure. What this literature and the measurement literature have not yet been made to say jointly is that a detector, a gate, or a score *is* a measurement model, with the same validity obligations as the fairness metric it polices. Documentation and process interventions [18, 34, 37] share our premise that fairness work needs infrastructure; audit practice [36, 41, 42] and traceability [28] give the released-artifact design its rationale. The stakeholder line [14, 26, 31–33, 46] asks who should choose the criteria; our contribution is adjacent, showing that the apparatus resolves several such questions before anyone is asked, at layers where nobody thinks to look. Zamfirescu-Pereira et al. [50] diagnose human error inside an analysis pipeline; we diagnose the pipeline's own rulers.

3 METHODS

3.1 Settings and data

Two regulated lending decisions and one instrument-test setting. German Credit [21] stands in for consumer installment credit: 1,000 applications of 1990s German data, a structural analogue from a different market and era (§7). HMDA Georgia [10] is residential mortgage origination: 10,978 applications after population construction (below). The third setting, a Lending Club default task, contains *no protected class*: its `gender` column is synthetic, assigned by an independent coin flip (`np.random.random(n) > 0.5`). We retain it because a harness must be tested somewhere its verdicts are known to be vacuous, and two of this paper's instrument findings (§4.4) surfaced there; nothing in this paper treats it as a lending sector. Lending discrimination itself is documented across this span [3, 7, 15, 40, 43].

3.2 Deterministic metric layer

A pipeline computes disparate impact with AIF360 [4], cross-checked against Fairlearn [5], and assigns severity labels by a fixed rule. The per-attribute summary the downstream layers receive carries three fields (`DisparateImpact`, `TheilIndex`, `BiasFlag`), of which the Theil index is computed on the prediction vector alone and is therefore identical across attributes within a dataset; the per-dataset CSV carries difference metrics (demographic parity, equal opportunity, average odds) whose full-population values appear in appendix F's artifacts.[1] Population, not sample: every applicant is scored out-of-fold by stratified five-fold cross-validation (random forest, accuracy 0.8002, AUC 0.8018 on HMDA), so audits run over all 10,978 and 1,000 rows rather than a held-out split.

3.3 LLM interpretation layer

A language model interprets the computed metrics and never recomputes one; scoring is against the deterministic label, never another model's opinion, with one disclosed exception (§4.4.5). Track Q

[1] The Lending Club summary uses a different five-field schema than the two regulated settings, a dual-schema defect we report rather than repair, since repairing it would churn every downstream artifact.

Table 1: The instance inventory. Each row is one measurement in this audit where the construct operationalised is not the construct named. Every number is regenerable; appendix F maps each to its artifact and command.

Construct named	Operationalised as	The gap	The clean output	Where
Layer one: the metric library (verified in AIF360 0.6.1 / Fairlearn 0.13.0)				
"favorable outcome"	`favorable_label=1.0`, a constructor default	on a `loan_default` target, defaulting becomes the good outcome	DI = 1.2072 (disadvantaged group reported as favoured); oriented 0.9862	§4.1
"disparate impact"	a bare point ratio	no interval, n, or reference group ships with the number	0.6734 at n=31 prints as authoritatively as 0.9604 at n=4,019	§4.1, §4.3.1
"compared to whom"	highest-rate cell vs. designated control, unrecorded	the convention moves every ratio 3–4 points	0.6734 or 0.7012; 0.8632 or 0.8988	§4.2.2
Layer two: the documented construct (definitions the data and their publishers supply)				
"age" (protected class)	banded `age_group` audited at 40+	public age bands straddle 62, the boundary the data's own flag encodes; young and senior pool and cancel	0.9791, severity LOW (62+ cut: 0.8897)	§4.2.1
"discrimination"	outcome-rate disparity	silent about the model's inputs entirely	clean DI = 0.8888 from a model taking sex and age as inputs	§4.2.3
Layer three: the audit's own instruments (our released harness; evidence about building an audit)				
"refusal"	any narrative field under 40 characters	instruction-following terseness is flagged; API failures conflated with model behaviour	6 of 9 flagged with zero refusals; a provider blocked at a measured 19.79	§4.4.1
"fabrication"	a float literal absent from context	derivable arithmetic (0.8888 → 88.88%) counts as fabrication	the model that shows its work flags 17 vs. 4	§4.4.2
"mitigation"	a canonical algorithm name	where the problem is data scarcity, "collect more data" is the only defensible answer	the gate fails exactly those two attributes	§4.4.3
"severity"	CRITICAL below 0.72 = 0.80 × 0.9	the boundary has no statutory or statistical provenance	authoritative-looking band labels	§4.4.4
"capability"	mean score over heterogeneous tasks	the mean tracks task difficulty; the scorer penalises terse output	75% agreement decomposing to 33.3/91.7/100; a 14.7-point "prompt sensitivity" spread that is 6.7 without the terse cycle	§4.4.5

issues four frozen prompt strategies per (setting, attribute) pair and scores the returned severity against the deterministic label; Track H produces narrative audits judged against mechanical checks. Prompt builders, detectors, severity rule and rubric are printed verbatim in appendix E; a no-key replay path reproduces all scoring.

3.4 Population construction

The HMDA file begins at 20,000 rows and the audit runs on 10,978: an attrition of 45.1% that includes dropping `Race Not Available` (4,140 rows, 20.7% of the file), joint applications, and free-form race text. That is selection on the protected attribute, performed before any metric runs; §5 discusses it and appendix D prints the rule-by-rule table with the disparate-impact ratios recomputed with the dropped categories retained. HMDA's public data carry no credit score, the primary underwriting variable.

3.5 The four-fifths screen

One convention in this paper is itself an imported construct. The four-fifths screen originated in US employment guidance, not in credit; we retain 0.80 because it is the de facto screening convention in fairness practice, we use it only as a screen, and we note that the guidance's own text contains the caveat our §4.3 quantifies: greater differences "may not constitute adverse impact where the differences are based on small numbers and are not statistically significant."

4 RESULTS

4.1 Layer one: library defaults

Construct named: "favorable outcome," "disparate impact." Operationalised as: a constructor default of 1.0*; a bare point ratio.*

We inspected AIF360 0.6.1 and Fairlearn 0.13.0 directly; the claims in this section are about those libraries as verified, not about the field.

Which outcome is good is a default. AIF360's `BinaryLabelDataset` takes `favorable_label` with a default of 1.0. Credit risk targets are commonly coded with the adverse event as 1 (default, delinquency, charge-off, fraud), so on a column named `loan_default` the default asserts that defaulting is the good outcome. The minimal reproduction (appendix B; deterministic, pinned versions) returns DI = 1.2072 under the library default, with the disadvantaged group reported as *favoured*, and 0.9862 under correct orientation. This is one constructor in a two-API library. AIF360's other entry point, StandardDataset, makes favorable_classes a required argument and cannot be constructed without answering the question. Fairlearn's demographic_parity_ratio exposes no favorable-label parameter at all; it computes a min/max ratio of the prediction column *as encoded*, so on the same table it reports 0.8284, a ratio of default rates and a number about the wrong outcome entirely, and delegates the orientation question to whoever encoded the column, without recording that it did so.

The in-vivo version ran end to end in our own harness before we caught it: selection rates computed on predicted default gave the Lending Club synthetic gender attribute DI = 0.0 and a CRITICAL label, the most severe output the pipeline can produce, where the oriented value is 0.9931, near parity. Nothing was misused; AIF360 computed what it was asked. A target named approved makes label 1 favorable, a target named loan_default inverts it, and nothing in the metric signature records which case holds. It is now a required argument and a regression test in our pipeline, which converts a silent inversion into a build failure.

No uncertainty ships with a disaggregated ratio. Neither library attaches an interval or a standard error to a disparate-impact ratio, and neither reports the *n* it was computed on unless asked (Fairlearn offers count as an opt-in metric). A ratio resting on 31 people prints with the same four decimal places as one resting on 4,019. This is the defensible core of what practitioners experience as the "minimum subgroup size" problem: a floor is arguably not a metric library's job, but uncertainty is what the estimate *means*, and §4.3 measures what its absence costs.

Nothing records the comparison. A disparate-impact ratio has a denominator, and neither library's output records which group supplied it. §4.2.2 shows the choice moves every number in our tables.

4.2 Layer two: the documented construct

4.2.1 The protected-class boundary. Construct named: age, as the lending domain defines the protected boundary. Operationalised as: a banded age_group *attribute audited at* 40+.

Our pipeline audited age at 40+, on the strength of a code comment asserting that the threshold aligned with the domain's definition. It does not: 40+ is a threshold imported from employment practice. The boundary lending's own documentation defines is 62+, and the public mortgage data ship a dedicated applicant_age_above_62 flag for it.

The substitution is not cosmetic. Our banded age attribute pools young and senior applicants (*n*=2,304 and 3,167) against the middle band; the two move in opposite directions, so pooling cancels them. The banded attribute returns DI = 0.9791, severity LOW: a clean bill of health. Audited on the documented cut, the same model over the same 10,978 rows returns 0.8897, the largest age disparity in the dataset and the lowest of its four protected attributes. The flag required was present and unused in the raw HMDA file, and it is necessary rather than convenient: the public age bands straddle 62 (the "55–64" band cannot be split), so *no* banded attribute can represent the boundary. 0.8897 clears the 0.80 screen and the pipeline's BiasFlag remains false on both cuts: the boundary changes which question the number answers, not the verdict on this data.

4.2.2 The comparator. Construct named: disparity against the favoured. Operationalised as: a ratio against an unrecorded, convention-chosen denominator.

Every ratio in this paper's tables is measured, by our pipeline's convention, against the most-favoured adequately-sized cell, and no metric signature records that. Measured instead against a designated control group (White × Male), the same cells read differently: American Indian × Male 0.7012 rather than 0.6734, Black × Female 0.8988 rather than 0.8632. The convention moves every number in the table by three to four points (appendix A prints all cells under three conventions), and §4.3.2 shows the choice also changes how often an audit is *wrong*, because a data-dependent reference is itself re-estimated in every sample.

4.2.3 The inputs the metric cannot see. Construct named: discrimination. Operationalised as: outcome-rate disparity.

The German Credit model takes applicant sex and age as input features (direct use of the protected attributes, among the first questions a fairness review of a lending model would ask), and our pipeline ran that specification end to end and emitted a clean DI = 0.8888 without comment. We previously filed this under limitations; it is a result. An outcome-rate metric is a function of predictions and group labels alone: it is structurally silent about what the model consumes, so the cleanest possible metric output is compatible with the most direct possible use of a protected attribute. No disparity ratio, at any sample size, under any convention, could have flagged it. Input scrutiny and outcome scrutiny are different measurements [19], and a toolchain that computes only the second says nothing about the first.

4.3 Measurement validity

4.3.1 Intervals on the real audit. Our audit originally ran on a held-out split of 2,196 mortgage applications and reported, among other things, that American Indian men were the *best*-treated group in the data (DI = 1.1504), on the evidence of four applicants, who all happened to be approved. Over the full population the same group is the worst cell in the dataset: 0.6734 at *n*=31. The audit did not miss the disparity; it *inverted* it, and attached a reassuring number to the inversion. The floor was right about what it suppressed: American Indian × Female read 0.1917 on the split (one approval in six), which looks like the most severe disparity in the study and is noise: 0.8815 at *n*=25 in full. Small-group estimates are unreliable in both directions, and every affected cell belonged to a racial minority, which is mechanism rather than coincidence: a floor on group size is a floor on group population, so the groups whose treatment is least certain are exactly the groups least likely to be counted.

Table 2 prints the full-population race×sex table, in full because this paper's argument is precisely that partial tables mislead, with the Wilson-interval disparate-impact bounds beside every ratio (reference: Asian × Female, *n*=254, the pipeline's most-favoured-adequate-cell convention; floor 15). Six of twelve cells have no determinate severity band: their intervals straddle a band boundary, so the audit does not know which band they belong to. American Indian × Male, the lowest point estimate, has interval [0.453, 0.904]: consistent with a critical violation and with clearing the screen. It cannot be called the worst-treated group.

Table 2: HMDA `race`×`sex`, full population (10,978), out-of-fold predictions. Reference `Asian × Female` (n=254); conservative Wilson-based DI intervals; "band det." marks cells whose whole interval sits in one severity band. Generated by `scripts/interval_estimates.py`; appendix A repeats this table under two further reference conventions.

cell	n	rate	DI	95% CI	band det.
American Indian x Male	31	0.5806	0.6734	[0.453, 0.904]	–
Multiracial x Female	18	0.6111	0.7088	[0.429, 0.979]	–
Multiracial x Male	29	0.6552	0.7599	[0.526, 0.983]	–
Pacific Islander x Female	12	0.6667	0.7732	[0.434, 1.058]	–
Pacific Islander x Male	15	0.7333	0.8505	[0.534, 1.094]	–
Black x Female	1,994	0.7442	0.8632	[0.806, 0.937]	yes
Black x Male	1,707	0.7452	0.8643	[0.805, 0.940]	yes
American Indian x Female	25	0.7600	0.8815	[0.629, 1.087]	–
White x Female	2,318	0.7865	0.9121	[0.856, 0.986]	yes
White x Male	4,019	0.8281	0.9604	[0.907, 1.031]	yes
Asian x Male	556	0.8525	0.9888	[0.913, 1.080]	yes
Asian x Female	254	0.8622	1.0000	[0.906, 1.104]	yes

What survives is narrower and firmer: `Black × Female` (0.8632, [0.806, 0.937], n=1,994) and `Black × Male` (0.8643, [0.805, 0.940], n=1,707) have intervals wholly below parity. What this dataset supports is a precisely estimated prediction-rate disparity for the two Black cells, 0.86–0.90 depending on reference convention and wholly below parity under both, which a point-estimate ranking demotes to sixth and seventh place, behind five cells whose ordering is noise. Reporting a ratio to four decimals without an interval is what makes that demotion possible [8].

4.3.2 The subsampling study. The anecdote above is one draw from a distribution, so we measured the distribution. Treating the full-population table as ground truth θ, we drew B=2,000 simple random subsamples at each audit size $m \in \{500, 1{,}000, 2{,}196, 4{,}000, 8{,}000\}$ and ran five reporting policies on every draw: a floor at $n \geq 15$ (the pipeline's), a floor at $n \geq 30$ (which our own codebase also contained, in an adjacent module), no-floor point estimates, no-floor Wilson intervals that assert a verdict only when the interval clears or fails the screen entirely, and empirical-Bayes shrinkage toward the pooled rate. Each policy runs under three reference conventions, with θ defined per convention. We record how often each policy asserts a population-violating cell clear (false clearance), asserts a clearing cell in violation (false alarm), places a reported estimate on the wrong side of parity (sign inversion), and names the wrong worst-treated cell or declines to name one. Appendix C gives the full grid, the estimator specification, and the design notes; Figure 1 and Table 3 give the primary convention.

Three results order the section. First, at the audit size we actually used, *every policy that reports small cells is usually wrong*: no-floor point estimates falsely clear a violating cell in 83.8% of draws and put at least one reported estimate on the wrong side of parity in 79.0%. Our 1.1504 was not bad luck. Second, the floors never false-clear at this size for the least reassuring possible reason: they never report the violating cells at all. Both floors suppress six of twelve cells on average and misidentify the worst-treated group in 100% of draws; and the $n \geq 15$ floor starts leaking at m=4,000 (2.9%, reaching 26.3% at m=8,000), exactly when the violating cells begin to cross it with noisy estimates. A floor does not solve the small-cell problem; it postpones it to a larger audit and converts it to silence in the meantime. Third, the two policies that respect uncertainty fail in opposite, honest ways: the Wilson policy never false-clears and almost never misranks (6.0%), but abstains from naming a worst cell in 90.3% of draws (at full population its interval on the worst cell still straddles the screen), while empirical-Bayes shrinkage is the *most* confident false-clearer of all (100.0% at m=2,196, already 98.7% at m=500), because a prior centered on the pooled rate absorbs exactly the small cells where the disparities live. Floors trade false clearance for silence; intervals trade decisiveness for honesty; shrinkage trades variance for a bias that points, in this data, toward clearance. The choice among them is a policy about who becomes visible, and nothing in a reported ratio records which policy produced it.

4.4 Layer three: audit instruments

The findings in this section are about our released harness, measured on frozen artifacts, and are evidence about *building* an audit rather than about the field. Appendix E prints every implementation verbatim.

4.4.1 "Refusal" := fewer than 40 characters. Construct named: refusal to answer [48]. Operationalised as: any narrative field under 40 characters.

The detector flagged six of nine responses under the terse-output prompt strategy. None was a refusal: no flagged record contained a refusal phrase, a missing field, or an invalid severity. The detector treats any narrative field under 40 characters as effectively empty, so `how_to_fix: "DisparateImpactRemover"` (correct, on-spec, and naming the applicable post-processor [16] in 22 characters) registered as a refusal. Compliance with the instruction was the trigger. Reporting the raw output would have yielded a 66.7% refusal rate and a tidy story about constrained prompting inducing evasion, entirely an artifact of our instrument. (Three of the six flags are Lending Club records: the harness failed identically on the setting whose verdicts are vacuous, which is what that setting is for.)

Then it set policy. Applied to a second provider, the same detector flagged 19 of 24 cycles. Reanalysis of the frozen records (appendix E) separates what the flag conflated: 18 of the 24 records are *empty API results* (transport failures, not model behaviour), of which 14 scored zero and four scored 10.0 despite containing nothing, because the cause-alignment subscore returns a flat 10.0 whenever the deterministic baseline lists no cause labels. The provider's mean is 19.79 over all records; 47.50 excluding zero scores (the figure we previously reported, which still contains the four empty-but-scored records); and 72.50 over the six records that contain any content at all. One of those six, a terse correct answer scoring 85, is the only content-bearing record the refusal detector flagged. On the strength of the 19.79 figure, that provider was recorded as permanently blocked in our guardrail configuration, where the block persists.

4.4.2 "Fabrication" := a float literal absent from context. Construct named: asserting numbers that were never provided. Operationalised as: any decimal literal not string-matching the context.

Replaying identical packs against two model generations, `gpt-5-mini` trips 17 flags across 28 calls where `gpt-4o` trips 4. Read naively, the

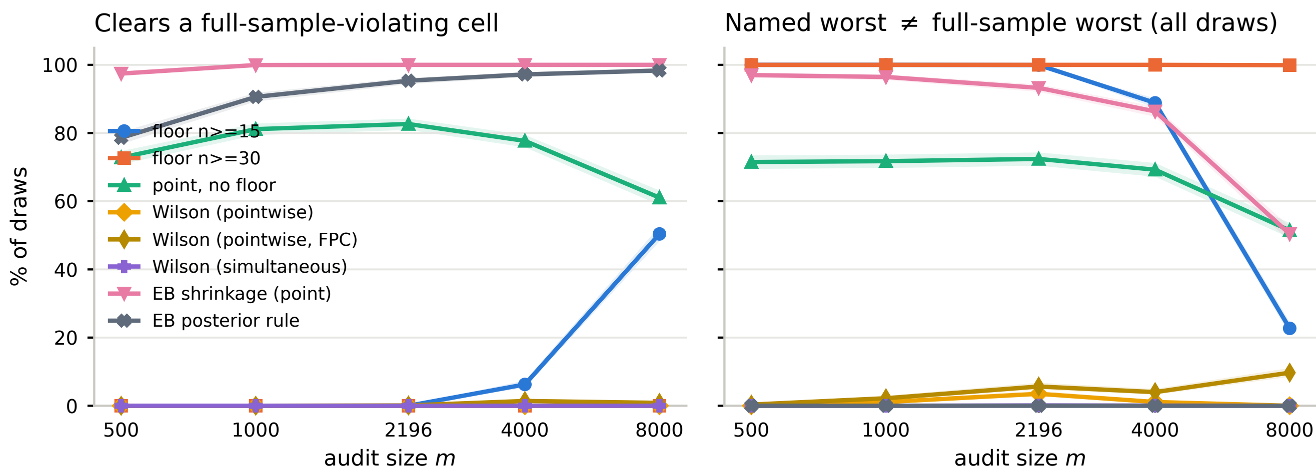

Figure 1: Error rates by audit size m (B=2,000 draws per size, shaded bands are 95% Wilson intervals on the rate; primary reference convention). Left: how often each policy asserts a population-violating cell clear of the 0.80 screen. The floor and interval policies coincide at zero for $m \leq 2{,}196$ (floors by silence, intervals by abstention), and the $n \geq 15$ floor begins to leak exactly when the violating cells cross it (2.9% at m=4,000, 26.3% at m=8,000). Right: how often the policy names the wrong worst-treated cell, given that it names one.

Table 3: All five policies at m=2,196, the size of the audit we originally ran. Rates are % of draws with 95% Wilson intervals; "abstains" is the share of draws where the policy declines to name a worst cell; "cells suppressed" is the mean count of the twelve `race`×`sex` cells not reported at all.

policy	false clearance % of draws	worst cell wrong % of draws	abstains %	sign inversion % of draws	false alarm % of draws	cells suppressed
floor n≥15	0.0 [0.0, 0.2]	100.0 [99.7, 100.0]	0.0	0.0 [0.0, 0.2]	17.9 [16.3, 19.6]	6.0
floor n≥30	0.0 [0.0, 0.2]	100.0 [99.8, 100.0]	0.0	0.0 [0.0, 0.2]	17.9 [16.3, 19.6]	6.0
point, no floor	83.8 [82.1, 85.3]	72.4 [70.4, 74.3]	0.0	79.0 [77.1, 80.7]	70.5 [68.5, 72.5]	0.1
Wilson interval	0.0 [0.0, 0.2]	6.0 [5.1, 7.2]	90.3	0.0 [0.0, 0.2]	0.6 [0.3, 1.1]	0.1
EB shrinkage	100.0 [99.8, 100.0]	93.2 [92.1, 94.3]	0.0	0.0 [0.0, 0.2]	0.4 [0.2, 0.8]	0.1

newer model fabricates four times as often. Checking each flagged value against the context, 14 of 21 (67%) are derivable from it: 88.88 is the disparate impact 0.8888 restated as a percentage, 83.87 is 0.8387, and most of the remainder are gaps between two provided values expressed in percentage points. The detector matches raw float literals, so converting a ratio to a percentage or subtracting two given numbers registers as fabrication. The model that reasons more explicitly is ranked as the less trustworthy one, precisely because it shows its work. The seven values that are not derivable (19.9 three times, 5.3 four times) remain honest fabrication candidates, and we report them as such.

4.4.3 *"Mitigation" := a canonical algorithm name. Construct named: a specific, actionable remedy. Operationalised as: a keyword match against a list of algorithm names.*

A guardrail requires every proposed mitigation to name a canonical algorithm: reweighing [24], disparate impact removal [16], and similar. It failed exactly the two attributes whose problem *is* data scarcity, where the audit proposed acquiring records to a stated floor, hierarchical shrinkage, and manual review. No post-processor repairs a six-record subgroup. The gate encodes an assumption that every fairness problem has an algorithmic mitigation, and where that is false it penalises the only defensible answer.

4.4.4 *Our own thresholds, same class. Construct named: severity. Operationalised as: fixed cuts at 0.72 and 0.80.*

The pipeline's severity rule places its CRITICAL boundary at 0.72. That number is 0.80×0.9, constructed in this project, with no external provenance, regulatory or statistical; it looks authoritative for the same reason the mis-sourced 40+ code comment did. The same codebase carried two reporting floors, $n \geq 15$ and $n \geq 30$, in adjacent modules, so the same cell was reportable or not depending on which module read it; §4.3.2 shows those two floors behave identically at small m and diverge exactly at the audit sizes where the $n \geq 15$ floor starts to leak. This is why the analysis in §4.3 uses only the labeled 0.80 screen plus intervals, and quotes the pipeline's band labels as pipeline convention rather than as findings. The code is left as implemented.

4.4.5 *Scores that operationalise "capability". Construct named: how good the model layer is. Operationalised as: mean score over heterogeneous tasks, graded by mechanical subscores.*

An agreement mean tracks the difficulty mix. Severity agreement between `gpt-4o` and the deterministic labels is 27/36 = 75.0%. It decomposes to 33.3%, 91.7% and 100.0% across the three settings, and the ordering tracks how discriminating each audited classifier is rather than how capable the model is: the 100% case is the setting where the classifier barely separates anyone, so every attribute is a trivial near-parity LOW and agreeing costs nothing. A single headline agreement figure for an LLM auditor is not interpretable apart from its difficulty mix. Errors were asymmetric in the safe direction (seven over-escalations, two under-escalations), which matters because under-escalation is the consequential failure for an audit tool.

Track H, disclosed. The narrative-audit track's rubric scores are one model's opinion of another's work (`gpt-4o` grading narratives generated by `o3`), which is exactly what this paper's architecture forbids: scoring against the deterministic label, never another model's opinion. We exclude those scores (4.8, 4.7, 5.0 of 5) rather than report them; an LLM-judged "quality" score is itself an unvalidated operationalisation [51], and this one was ours. What survives on mechanical evidence: all 27 of 27 citations in the narrative audits verified against the retrieved evidence pool. Two behaviours worth reporting as observations rather than scores: on a six-record subgroup the narrative audit independently reproduced the small-n caveat and quantified it (one label change would swing the ratio below the screen), and on the near-degenerate classifier it declined the flattering reading.

Prompt sensitivity, reanalysed. A widely reported result about LLM evaluators is that prompt phrasing moves the verdict [44]. It reproduced here with a generational asymmetry: across four prompt strategies on identical packs, the spread in cycle-mean score is 14.7 and 9.0 points for `gpt-4o` (German Credit, HMDA) against 0.0 and 1.9 for `gpt-5-mini`. We previously reported that contrast as model-generational. Reanalysis of the frozen records (appendix E) shows the refusal flag never feeds the score, so no refusal penalty inflates it. But the `gpt-4o` spread is concentrated in the one strategy that demands terse output: excluding the constrained cycle it falls to 6.7 and 1.25, and the cycle-four dip decomposes almost entirely into the completeness and severity-agreement subscores (78%/22% on German Credit, 42%/58% on HMDA), subscores that reward enumerating content, scored on answers instructed to be short. Every `gpt-4o` constrained-cycle record was simultaneously flagged by the 40-character refusal rule; the scorer and the detector were measuring the same thing, brevity, under two different construct names. A generational difference survives (`gpt-5-mini`'s spread is near zero under every treatment), but the published magnitude was partly our instrument, and paraphrase-instability claims should carry a model, a date, *and* an instrument audit. (`gpt-5-mini` also rejects the `temperature` parameter outright, so a repeated-sampling study is impossible on it, an API constraint that silently determines which experiments a benchmark can run.)

5 DISCUSSION

The layer-one findings hold for anyone using the verified libraries; the rest are about choices our audit made because nothing made them for us, and that is the point. An audit's numbers mean what its readers take them to mean only if the construct questions of Table 1 are answered correctly. The toolchain answers the favorable-outcome question by default and the rest not at all, so each is settled privately, by whoever writes the harness, without a record.

The errors have a direction. A false alarm costs reviewer time; a false clearance closes an open question. Nearly every gap in Table 1 produced its error in the clearance direction: the banded age attribute read LOW, the small-sample audit read its worst cell as its best, the shrinkage policy cleared violating cells with the most confidence of any policy, and the "prompt sensitivity" figure flattered the older model's critics rather than its scorer. Clean results were the outputs we were least likely to re-examine, and every mechanism in this paper pushes toward clean.

The instruments make it worse, because every mechanical proxy for judgement is a policy. A length threshold that treats brevity as refusal is a policy about what counts as an answer. A keyword list that treats "acquire more data" as a non-mitigation is a policy about what counts as a remedy. A float-literal match that treats a percentage as a fabrication is a policy about what counts as evidence. None was labelled a policy; each was a heuristic written to make an evaluation automatable, and several punished the better answer in the cases where judgement was most needed. As evaluation automates further (LLM judges, automated red-teams, compliance scoring), this class of failure has room to scale with it.

Two of our own decisions belong in this catalogue. Our population construction dropped 45.1% of the raw HMDA file before any metric ran, including every applicant whose race is not available (20.7% of the file): selection on the protected attribute, deciding who counts as protected upstream of every number in this paper. Appendix D prints what the dropped categories look like when retained (the excluded `Race Not Available` group approves at 0.7234, below every retained majority group). And our division of labour (deterministic code owns every number, the model owns only interpretation, scoring runs against the numbers) protected every *number* in this paper and not one of the *operationalisations*: it is the reason we can state which of our results were artifacts, and it is no defence against choosing the wrong construct in the first place.

6 RECOMMENDATIONS

Each item follows from a measured failure above.

(1) **State the favorable outcome as an explicit, required argument.** Never infer it from label values (§4.1).
(2) **Report the interval, the *n*, and the reference group beside every ratio**, and print what any floor withheld: suppressed cells with their n and their unreported value. The audit of §4.3.1 was not wrong about any number it computed; it was wrong about what it did not print.
(3) **Bind protected-class definitions to their documented source** in code, with the citation next to the threshold. A comment asserting the wrong provenance survived review precisely because it looked authoritative (§4.2.1).

(4) **Treat every mechanical gate as a hypothesis to falsify**, starting with inputs where the correct answer is unusual: the terse answer, the tiny subgroup, the model that shows its work (§4.4).
(5) **Audit the population, not a convenience sample, and disclose population construction.** Most of our small-cell instability came from auditing a split; all of our attrition happened before any metric ran (§4.3.2, appendix D).
(6) **Name the reporting policy.** Floor, interval, or shrinkage is a choice among failure modes, and §4.3.2 shows they disagree massively at realistic audit sizes. A reported table should say which policy produced it.

These steps are necessary, not sufficient. The list is what our measured failures imply, not an enumeration of the ways an audit can be wrong, and completing it does not make a model fair or a lender compliant. A real review involves far more than a disparity screen; we perform the screening step only. Anyone citing this protocol as evidence of compliance is misusing it.

7 LIMITATIONS

Scale. Two lending datasets, one geography and era each, two primary models. These are mechanism-level findings: we show that a class of failure occurs and how it operates, not how often it occurs in the wild. The German data are a structural analogue from a different market and era, and no US conclusion is drawn from them.

No human subjects. We make claims about what an audit reader would conclude without testing them on audit readers. That is the natural next study, and the main gap between this work and the stakeholder line it builds on [31, 32].

No causal or legal claims. All metrics are associational; root-cause mapping is rule-based inference over metric values. Regulation appears in this paper only as the provenance of definitions and flags, never as something we interpret, and we treat a disparate-impact ratio below 0.80 as a screening signal only, drawing no conclusion about discrimination [2, 47].

Model limitations. The German Credit classifier uses protected attributes as features (reported as a result, §4.2.3); the Lending Club protected attribute is synthetic; HMDA lacks credit score. Each bounds what the corresponding audit can mean.

Sampling confound. Temperature and seed are unpinned in the interpretation layer, so within-model cycle differences mix prompt strategy with sampling noise; the cross-model contrast in §4.4.5 is robust to this only because both models face the same confound, and the instrument decomposition there is the reason we no longer quote the within-model spread as a finding. An earlier ungrounded-vs-grounded citation comparison straddles a pipeline revision and is not quoted anywhere in this paper because it is a legacy contrast, not an ablation.

The subsampling study measures our pipeline's conventions. Ground truth is the full-population table under a stated reference convention; different conventions define different θ, which is part of the finding but also a limit on transportability. The DI intervals are deliberately conservative and over-cover (0.993–1.0 measured against nominal 0.95); appendix C details this and the estimator choices.

8 CONCLUSION

At the layer of the library, a constructor default decided which outcome of a loan is good. At the layer of the definition, an attribute banding decided which age question the audit asked, and the reporting policy decided whether a violating subgroup was falsely cleared, silently omitted, or honestly left undetermined. At the layer of the audit's own instruments, a 40-character rule decided what counted as a refusal, a float match decided what counted as fabrication, and two mechanical subscores account for much of a "prompt sensitivity" result we ourselves previously reported.

Every one of these is the same event: an intended, documented construct, operationalised as something adjacent, producing a clean number, recorded nowhere. The gap is not closable in general, because every measurement operationalises, but it is *recordable*, and §6 is the record this audit would have needed: the outcome, the interval, the n, the reference group, the definition's source, the reporting policy, and an adversarial test for every gate. None of it is technically difficult. All of it is currently optional, which is the problem.

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A FULL CELL TABLES UNDER THREE REFERENCE CONVENTIONS

All tables are generated by `scripts/interval_estimates.py --emit-tex` from the out-of-fold prediction files; nothing is transcribed by hand. The conventions: *max rate above floor* (the pipeline's rule; reference is the highest-rate cell with $n \geq 15$), *max rate* (no floor), and *control* (`White × Male` for HMDA, `40_plus × male` for German Credit). The body's Table 2 is the first of these.

HMDA race × sex, reference = max rate (no floor)

cell	n	rate	DI	95% CI	band det.
American Indian x Male	31	0.5806	0.6734	[0.453, 0.904]	–
Multiracial x Female	18	0.6111	0.7088	[0.429, 0.979]	–
Multiracial x Male	29	0.6552	0.7599	[0.526, 0.983]	–
Pacific Islander x Female	12	0.6667	0.7732	[0.434, 1.058]	–
Pacific Islander x Male	15	0.7333	0.8505	[0.534, 1.094]	–
Black x Female	1,994	0.7442	0.8632	[0.806, 0.937]	yes
Black x Male	1,707	0.7452	0.8643	[0.805, 0.940]	yes
American Indian x Female	25	0.7600	0.8815	[0.629, 1.087]	–
White x Female	2,318	0.7865	0.9121	[0.856, 0.986]	yes
White x Male	4,019	0.8281	0.9604	[0.907, 1.031]	yes
Asian x Male	556	0.8525	0.9888	[0.913, 1.080]	yes
Asian x Female	254	0.8622	1.0000	[0.906, 1.104]	yes

HMDA race × sex, reference = White × Male

cell	n	rate	DI	95% CI	band det.
American Indian x Male	31	0.5806	0.7012	[0.486, 0.902]	–
Multiracial x Female	18	0.6111	0.7380	[0.460, 0.977]	–
Multiracial x Male	29	0.6552	0.7912	[0.564, 0.981]	–
Pacific Islander x Female	12	0.6667	0.8051	[0.465, 1.056]	–
Pacific Islander x Male	15	0.7333	0.8856	[0.572, 1.092]	–
Black x Female	1,994	0.7442	0.8988	[0.863, 0.935]	yes
Black x Male	1,707	0.7452	0.8999	[0.863, 0.938]	yes
American Indian x Female	25	0.7600	0.9178	[0.674, 1.085]	–
White x Female	2,318	0.7865	0.9497	[0.916, 0.984]	yes
White x Male	4,019	0.8281	1.0000	[0.972, 1.029]	yes
Asian x Male	556	0.8525	1.0295	[0.978, 1.078]	yes
Asian x Female	254	0.8622	1.0412	[0.970, 1.102]	yes

German Credit AgeGroup × Sex, reference = max rate above floor

cell	n	rate	DI	95% CI	band det.
under_40 x female	241	0.7012	0.8201	[0.697, 1.003]	–
under_40 x male	460	0.8217	0.9610	[0.853, 1.134]	yes
40_plus x male	230	0.8391	0.9814	[0.855, 1.169]	yes
40_plus x female	69	0.8551	1.0000	[0.820, 1.220]	yes

German Credit AgeGroup × Sex, reference = max rate (no floor)

cell	n	rate	DI	95% CI	band det.
under_40 x female	241	0.7012	0.8201	[0.697, 1.003]	–
under_40 x male	460	0.8217	0.9610	[0.853, 1.134]	yes
40_plus x male	230	0.8391	0.9814	[0.855, 1.169]	yes
40_plus x female	69	0.8551	1.0000	[0.820, 1.220]	yes

German Credit AgeGroup × Sex, reference = 40_plus × male

cell	n	rate	DI	95% CI	band det.
under_40 x female	241	0.7012	0.8357	[0.727, 0.961]	–
under_40 x male	460	0.8217	0.9793	[0.890, 1.086]	yes
40_plus x male	230	0.8391	1.0000	[0.892, 1.121]	yes
40_plus x female	69	0.8551	1.0190	[0.855, 1.169]	yes

Marginal disparate impact, full population, for completeness: German Credit Sex 0.8888, AgeGroup 0.9258, `foreign_worker` 0.8387; HMDA `race` 0.9371, `sex` 0.9572, `age_group` (banded) 0.9791, `age_62_plus` 0.8897; Lending Club (synthetic gender; instrument-test setting only) `gender` 0.9931, `income_level` 1.0072, `loan_amount_level` 1.0063. Source: `artifacts/*/fairness/fairness_summary.json`.

B MINIMAL REPRODUCTION OF THE FAVORABLE-LABEL DEFAULT

The script (deterministic: fixed counts, no RNG), its complete output, and the pinned environment (`requirements-repro.txt`: `aif360==0.6.1`, `fairlearn==0.13.0`). Two groups of 1,772; the privileged group defaults 111 times, the unprivileged 134.

```
#!/usr/bin/env python3.11
"""Minimal reproduction: AIF360's favorable-label default inverts a DI ratio.

Two groups of 1,772 applicants; the privileged group defaults 111 times, the
unprivileged 134. On a target named `loan_default`, `BinaryLabelDataset`
defaults `favorable_label=1.0` -- asserting that defaulting is the favorable
outcome -- and reports DI 1.2072 (the unprivileged group "favored"). Stating
`favorable_label=0.0` gives the oriented value, 0.9862. Both numbers come from
the same table; the sign of the conclusion is set by a constructor default.

Scope: this is one constructor in a two-API library. `StandardDataset`
requires `favorable_classes` and cannot be constructed without answering the
question. Fairlearn's `demographic_parity_ratio` exposes no favorable-label
parameter at all: it takes the min/max ratio of `y_pred`'s selection rate as
encoded, so on this table it reports 0.8284 -- a ratio of *default* rates,
a number about the wrong outcome -- and the orientation question is delegated
entirely to whoever encoded the column.

Install: pip install -r requirements-repro.txt
Run:     python3.11 scripts/aif360_default_repro.py
Output verbatim in artifacts/consolidated/aif360_default_repro.txt.
"""
import aif360
import fairlearn
import numpy as np
import pandas as pd
from aif360.datasets import BinaryLabelDataset
from aif360.metrics import BinaryLabelDatasetMetric
from fairlearn.metrics import demographic_parity_ratio

N, K_PRIV, K_UNPRIV = 1772, 111, 134  # defaults per group of N

df = pd.DataFrame(
    {
        "loan_default": [1] * K_PRIV + [0] * (N - K_PRIV)
        + [1] * K_UNPRIV + [0] * (N - K_UNPRIV),
        "group": [1] * N + [0] * N,  # 1 = privileged
    }
)
```

```
kwargs = dict(
    df=df, label_names=["loan_default"], protected_attribute_names=["group"]
)
groups = dict(privileged_groups=[{"group": 1}], unprivileged_groups=[{"group": 0}])

di_default = BinaryLabelDatasetMetric(
    BinaryLabelDataset(**kwargs), **groups  # favorable_label defaults to 1.0
).disparate_impact()
di_oriented = BinaryLabelDatasetMetric(
    BinaryLabelDataset(favorable_label=0.0, unfavorable_label=1.0, **kwargs),
    **groups,
).disparate_impact()
dpr_as_encoded = demographic_parity_ratio(
    y_true=df["loan_default"], y_pred=df["loan_default"], sensitive_features=df["group"]
)
dpr_oriented = demographic_parity_ratio(
    y_true=1 - df["loan_default"], y_pred=1 - df["loan_default"],
    sensitive_features=df["group"],
)

print(f"aif360=={aif360.__version__} fairlearn=={fairlearn.__version__}")
print(f"numpy=={np.__version__} pandas=={pd.__version__}")
print(f"default favorable_label=1.0 on 'loan_default': DI = {di_default:.4f}")
print(f"oriented favorable_label=0.0:                  DI = {di_oriented:.4f}")
print("fairlearn demographic_parity_ratio, column as encoded:")
print(f"  {dpr_as_encoded:.4f} (a min/max ratio of default rates)")
print("fairlearn demographic_parity_ratio, column re-encoded by hand:")
print(f"  {dpr_oriented:.4f} (no favorable-label parameter exists)")

assert round(di_default, 4) == 1.2072
assert round(di_oriented, 4) == 0.9862
assert round(dpr_as_encoded, 4) == 0.8284  # a ratio about the wrong outcome
assert round(dpr_oriented, 4) == 0.9862
```

Output (`artifacts/consolidated/aif360_default_repro.txt`):

```
aif360==0.6.1 fairlearn==0.13.0
numpy==2.4.4 pandas==3.0.2
default favorable_label=1.0 on 'loan_default': DI = 1.2072
oriented favorable_label=0.0:                  DI = 0.9862
fairlearn demographic_parity_ratio, column as encoded:
  0.8284 (a min/max ratio of default rates)
fairlearn demographic_parity_ratio, column re-encoded by hand:
  0.9862 (no favorable-label parameter exists)
```

C SUBSAMPLING STUDY: DESIGN AND FULL GRID

Design. Population: the 10,978 out-of-fold HMDA predictions (random forest, stratified five-fold, seed 42). Draws: simple random, without replacement, B=2,000 per size, seeded; note the historical 2,196 split was outcome-stratified, so the study's draws are a slightly harder regime than the original audit faced. Ground truth θ is the full-population cell table *under each reference convention*; violating cells are those with θ-DI below 0.80 excluding the reference (four cells under the max-rate conventions, three under the control convention). Policies as in §4.3.2. The Wilson policy's DI interval is deliberately conservative (disadvantaged cell's lower bound over the reference's upper bound, ignoring the correlation between the two estimates) and it over-covers: measured coverage 0.993–1.0 against nominal 0.95. That is the intended direction of error for an audit, and it is why the abstention rates in the grid are upper bounds on what a sharper interval (delta method, bootstrap) would produce [8]. The empirical-Bayes policy shrinks cell rates toward the pooled rate with a beta prior whose mean is the pooled rate and whose strength $\tau = (1 - \rho)/\rho$ comes from the weighted (Kleinman) beta-binomial moment estimator of the intraclass correlation ρ, clipped to $[10^{-4}, 0.99]$; on this population $\tau \approx 90$, which is why the policy clears small violating cells even at full population strength. A data-dependent reference (both max-rate conventions) is re-fit within every draw, because that is what the convention *is*, while the control reference is fixed.

Full grid. Rates in % of draws.

Reference convention: max rate.

policy	m	false clr.	worst wrong	abstain	sign inv.
floor n≥15	500	0.0	100.0	0.0	0.0
	1000	0.0	100.0	0.0	0.0
	2196	0.0	100.0	0.0	0.0
	4000	2.0	88.8	0.0	0.0
	8000	25.1	22.7	0.0	0.0
floor n≥30	500	0.0	100.0	0.0	0.0
	1000	0.0	100.0	0.0	0.0
	2196	0.0	100.0	0.0	0.0
	4000	0.0	100.0	0.0	0.0
	8000	0.0	99.9	0.0	0.0
point, no floor	500	76.6	71.5	0.0	0.0
	1000	81.5	71.8	0.0	0.0
	2196	77.2	72.4	0.0	0.0
	4000	72.0	69.2	0.0	0.0
	8000	58.8	51.4	0.0	0.0
Wilson interval	500	0.0	0.2	99.6	0.0
	1000	0.0	0.5	99.1	0.0
	2196	0.0	1.7	97.5	0.0
	4000	0.0	1.8	96.6	0.0
	8000	0.0	0.1	98.6	0.0
EB shrinkage	500	98.6	97.0	0.0	0.0
	1000	100.0	96.5	0.0	0.0
	2196	100.0	93.2	0.0	0.0
	4000	100.0	86.4	0.0	0.0
	8000	100.0	50.4	0.0	0.0

Reference convention: max rate floor15.

policy	m	false clr.	worst wrong	abstain	sign inv.
floor n≥15	500	0.0	100.0	0.0	0.0
	1000	0.0	100.0	0.0	0.0
	2196	0.0	100.0	0.0	0.0
	4000	2.9	88.8	0.0	0.0
	8000	26.3	22.7	0.0	0.0
floor n≥30	500	0.0	100.0	0.0	0.0
	1000	0.0	100.0	0.0	0.0
	2196	0.0	100.0	0.0	0.0
	4000	0.0	100.0	0.0	0.0
	8000	0.0	99.9	0.0	0.0
point, no floor	500	77.6	71.5	0.0	88.0
	1000	83.8	71.8	0.0	90.6
	2196	83.8	72.4	0.0	79.0
	4000	78.3	69.2	0.0	49.6
	8000	60.0	51.4	0.0	8.2
Wilson interval	500	0.0	0.8	98.5	0.1
	1000	0.0	3.0	94.7	0.0
	2196	0.0	6.0	90.3	0.0
	4000	0.1	2.9	94.0	0.5
	8000	0.0	0.1	98.6	0.1
EB shrinkage	500	98.7	97.0	0.0	0.1
	1000	100.0	96.5	0.0	0.0
	2196	100.0	93.2	0.0	0.0
	4000	100.0	86.4	0.0	0.0
	8000	100.0	50.4	0.0	0.0

Reference convention: control.

policy	m	false clr.	worst wrong	abstain	sign inv.
floor n≥15	500	0.0	100.0	0.0	55.8
	1000	0.0	100.0	0.0	58.0
	2196	0.0	100.0	0.0	40.6
	4000	6.2	88.8	0.0	23.4
	8000	50.4	22.7	0.0	10.8
floor n≥30	500	0.0	100.0	0.0	29.8
	1000	0.0	100.0	0.0	41.9
	2196	0.0	100.0	0.0	40.6
	4000	0.0	100.0	0.0	23.2
	8000	0.0	99.9	0.0	0.9
point, no floor	500	72.8	71.5	0.0	97.6
	1000	81.2	71.8	0.0	97.5
	2196	82.7	72.4	0.0	91.0
	4000	77.7	69.2	0.0	74.0
	8000	61.1	51.4	0.0	25.0
Wilson interval	500	0.0	0.9	98.5	25.2
	1000	0.0	3.0	94.7	18.8
	2196	0.1	6.0	90.0	31.4
	4000	0.4	3.1	93.7	24.6
	8000	0.0	0.2	99.0	2.2
EB shrinkage	500	97.5	97.0	0.0	93.5
	1000	100.0	96.5	0.0	87.5
	2196	100.0	93.2	0.0	74.8
	4000	100.0	86.4	0.0	53.5
	8000	100.0	50.4	0.0	10.7

D POPULATION CONSTRUCTION

Generated by `scripts/hmda_attrition.py`, which replays `clean_hmda.py`'s row-dropping rules in order.

Attrition, 20,000 → 10,978

step	categories dropped (count)	dropped	remaining
raw file	–	0	20,000
race	Race Not Available (4140); Joint (280); Free Form Text Only (3)	4,423	15,577
sex	Joint (4529); Sex Not Available (69)	4,598	10,979
age	8888 (1)	1	10,978

Race DI on raw approval rates, dropped categories retained

Raw approvals (`action_taken == 1`) over all 20,000 rows: a property of the data, not the model, since dropped rows never receive predictions. The excluded `Race Not Available` group (4,140 applicants) approves at 0.7234: it clears the 0.80 screen against White applicants (0.9205) and sits below every retained majority group, and its exclusion is invisible in every downstream table of this paper.

race	*n*	approval rate	DI vs White	DI vs best
Joint (dropped)	280	0.8036	1.0225	1.0000
Asian	1,063	0.7959	1.0127	0.9904
White	9,822	0.7859	1.0000	0.9780
Race Not Available (dropped)	4,140	0.7234	0.9205	0.9003
Black	4,541	0.6410	0.8157	0.7977
Pacific Islander	32	0.5312	0.6760	0.6611
American Indian	63	0.5238	0.6665	0.6519
Multiracial	56	0.5000	0.6362	0.6222
Free Form Text Only (dropped)	3	0.0000	0.0000	0.0000

E THE INSTRUMENTS, VERBATIM

Everything in this appendix is extracted from the live code by `scripts/emit_appendix_f.py` (via `inspect.getsource`) and kept current by a fidelity test, so it cannot drift from what actually ran. The frozen prompt packs (fully rendered, with data) are in `artifacts/llm_benchmark/dry_run/`.

Cycle-4 decomposition backing §4.4.5

Generated by `scripts/recheck_prompt_sensitivity.py`. "Share of dip" columns decompose the constrained-cycle drop in mean total score by subscore.

			spread excl.	cycle-4	share of dip	
model	dataset	spread	constrained	dip	completeness	severity
gpt-4o	german credit	14.67	6.67	10.22	0.78	0.22
gpt-4o	hmda	9.00	1.25	8.58	0.42	0.58
gpt-5-mini	german credit	0.00	0.00	0.00	–	–
gpt-5-mini	hmda	1.88	1.88	0.62	0.00	0.00

Lines longer than 64 columns are wrapped with a trailing backslash for typesetting; the repository holds the byte-exact sources.

The four prompt strategies (one builder per cycle)

`scripts/llm_benchmark.py`

```
def build_zero_shot_prompt(
    context_payload: dict, attr: str, dataset_name: str, researc\
    h_ctx: str,
    **_kwargs,
) -> str:
    ctx = _single_attr_context(context_payload, attr)
    schema = _OUTPUT_SCHEMA.format(attr=attr)
    return textwrap.dedent(f"""\
    You are an AI fairness auditor. Python has pre-computed the \
    fairness metrics
    below for the "{dataset_name}" dataset. Do NOT recompute the\
     math.
    Provide qualitative analysis for the protected attribute "{a\
    ttr}".

    {AIF360_CONTEXT}

    ## Research evidence (Semantic Scholar)
    {research_ctx}

    ## Fairness context (pre-computed)
    ```json
    {ctx}
    ```

    {schema}
    """)
```

`scripts/llm_benchmark.py`

```
def build_chain_of_thought_prompt(
    context_payload: dict, attr: str, dataset_name: str, researc\
    h_ctx: str,
    **_kwargs,
) -> str:
    ctx = _single_attr_context(context_payload, attr)
    schema = _OUTPUT_SCHEMA.format(attr=attr)
    return textwrap.dedent(f"""\
    You are an AI fairness auditor. Reason step by step before w\
    riting your answer.

    Step 1 — Read each metric value for "{attr}" in "{dataset_na\
    me}".
    Step 2 — Identify which metric(s) signal the most severe dis\
    parity and why.
    Step 3 — Trace the disparity back to its structural or histo\
    rical cause.
    Step 4 — Select a mitigation algorithm from the AIF360/Fairl\
    earn list that
              specifically addresses that cause. Explain why it \
    fits.
    Step 5 — Assign severity using the DI threshold rules in the\
     context.
    Step 6 — Output your final answer as JSON only (no step-by-s\
    tep prose in output).

    {AIF360_CONTEXT}

    ## Research evidence (Semantic Scholar)
    {research_ctx}

    ## Fairness context (pre-computed)
    ```json
    {ctx}
    ```

    {schema}
    """)
```

`scripts/llm_benchmark.py`

```
def build_self_critique_prompt(
    context_payload: dict, attr: str, dataset_name: str, researc\
    h_ctx: str,
    previous_output: dict,
    **_kwargs,
) -> str:
    ctx = _single_attr_context(context_payload, attr)
    prev_json = json.dumps(previous_output, indent=2)
    schema = _OUTPUT_SCHEMA.format(attr=attr)
    return textwrap.dedent(f"""\
    You are an AI fairness auditor reviewing your own prior anal\
    ysis of "{attr}"
    in the "{dataset_name}" dataset.

    ## Your previous output
    ```json
    {prev_json}
    ```

    Self-critique checklist — assess each point, then improve:
```

```
    1. Severity: is it consistent with the DI threshold rules in\
     the context?
    2. Cause: is "why_is_wrong" specific (cites a mechanism) or \
    generic (says "bias")?
    3. Fix: does "how_to_fix" name a specific algorithm from AIF\
    360 or Fairlearn?
    4. Citations: are the research entries real and directly rel\
    evant?

    {AIF360_CONTEXT}

    ## Research evidence (Semantic Scholar)
    {research_ctx}

    ## Fairness context (pre-computed)
    ```json
    {ctx}
    ```

    {schema}
    """)
```

scripts/llm_benchmark.py

```
def build_constrained_prompt(
    context_payload: dict, attr: str, dataset_name: str, researc\
    h_ctx: str,
    **_kwargs,
) -> str:
    ctx = _single_attr_context(context_payload, attr)
    alg_list = "Reweighing, DisparateImpactRemover, Exponentiate\
    dGradient, ThresholdOptimizer"
    schema = _OUTPUT_SCHEMA.format(attr=attr)
    return textwrap.dedent(f"""\
    You are an AI fairness auditor operating under strict output\
     constraints.

    ## Hard constraints (violations invalidate the response)
    1. "what_is_wrong" MUST quote the exact numerical DI value f\
    rom the context.
    2. "how_to_fix" MUST name exactly one algorithm from: {alg_l\
    ist}.
    3. "severity" MUST be exactly one of: CRITICAL, HIGH, MODERA\
    TE, LOW — no other text.
    4. "supporting_research" MUST contain at least 2 entries.
    5. Do NOT speculate beyond what the provided metric values s\
    upport.

    ## Attribute: "{attr}"  |  Dataset: "{dataset_name}"

    {AIF360_CONTEXT}

    ## Research evidence (Semantic Scholar)
    {research_ctx}

    ## Fairness context (pre-computed)
    ```json
    {ctx}
    ```

    {schema}
    """)
```

The refusal detector (§7.1); the 40-character rule is the brevity operationalisation the paper reports

scripts/llm_benchmark.py

```
def detect_refusal(result: dict | None, raw_text: str = "") -> b\
    ool:
    """Return True if the LLM refused or gave an unusably short \
    answer."""
    if result is None:
        return True
    lower = raw_text.lower()
    if any(phrase in lower for phrase in REFUSAL_PHRASES):
        return True
    for field in REQUIRED_RESPONSE_FIELDS:
        if not result.get(field):
            return True
    # Flag very short paragraphs (< 40 chars) as effectively emp\
    ty
    for field in ("what_is_wrong", "why_is_wrong", "how_to_fix")\
    :
        if len(str(result.get(field, ""))) < 40:
            return True
    sev = str(result.get("severity", "")).upper().strip()
    if sev and sev not in VALID_SEVERITIES:
        return True
    return False
```

The fabrication detector (§7.2): float literals absent from context, citations without a matching title 3-gram

scripts/llm_benchmark.py

```
def known_metric_values(context_payload: dict) -> set[float]:
    """Collect every numeric value in the metric context (rounde\
    d for matching)."""
    values: set[float] = set()

    def walk(node: Any) -> None:
        if isinstance(node, dict):
            for v in node.values():
                walk(v)
        elif isinstance(node, list):
            for v in node:
                walk(v)
        elif isinstance(node, (int, float)) and not isinstance(n\
    ode, bool):
            values.add(float(node))

    walk(context_payload)
    return values
```

scripts/llm_benchmark.py

```
def detect_hallucinations(
    result: dict,
    known_titles: list[str],
    context_values: set[float] | None = None,
) -> list[str]:
    """
    Flag content not traceable to the provided context (guardrai\
    l 5:
    LLMs interpret metrics, they never assert new ones).

    Two heuristics:
    - Citations: suspicious if no 3-gram appears in any known pa\
    per title
      (case-insensitive); short single-word entries are also fla\
    gged.
    - Metric values: any decimal number in the narrative fields \
    that is not
      within tolerance of a value present in the metric context \
    (and is not a
      canonical threshold such as 0.8) is flagged as a fabricate\
    d metric.
    """
    if not result:
        return []

    flags: list[str] = []

    cited = result.get("supporting_research", [])
    known_blob = " ".join(t.lower() for t in known_titles)
    for citation in cited:
        words = citation.lower().split()
        if len(words) < 3:
            flags.append(citation)
            continue
        trigrams = [" ".join(words[i : i + 3]) for i in range(le\
    n(words) - 2)]
        if not any(tg in known_blob for tg in trigrams):
            flags.append(citation)

    if context_values is not None:
        allowed = context_values | _ALLOWED_METRIC_ANCHORS
        for field in ("what_is_wrong", "why_is_wrong", "how_to_f\
    ix"):
            for match in _METRIC_VALUE_RE.findall(str(result.get\
    (field, ""))):
                quoted = float(match)
                if not any(abs(quoted - v) <= _METRIC_TOLERANCE \
    for v in allowed):
                    flags.append(f"metric value {match} in '{fie\
    ld}' not present in provided context")

    return flags
```

The severity rule (§7.4); the CRITICAL boundary is 0.8 x 0.9 = 0.72, constructed here, with no external provenance

scripts/qualitative_analysis.py

```
def classify_severity(di: float | None, threshold: float = 0.8) \
    -> str:
    if di is None:
        return "UNKNOWN"
    if di < threshold * 0.9:        # e.g. < 0.72
        return "CRITICAL"
    if di < threshold:              # e.g. < 0.80
        return "HIGH"
    if di < 0.95:
        return "MODERATE (borderline)"
    return "LOW (fair)"
```

The scoring rubric (§7.5): weights 40/20/15/15/10; note cause-alignment's flat 10.0 when the baseline lists no cause labels – the default that scored four empty responses

scripts/llm_benchmark_common.py

```
def score_llm_output(result: dict[str, Any], baseline_payload: d\
    ict[str, Any]) -> dict[str, Any]:
    """Score a qualitative LLM output against the deterministic \
    baseline."""
    baseline_by_attr = {
        row["attribute"]: row for row in baseline_payload.get("a\
    ttributes", [])
    }
    result_by_attr = {
        row["attribute"]: row for row in result.get("qualitative\
    ", [])
    }

    completeness = _score_completeness(result, expected_attrs=li\
    st(baseline_by_attr.keys()))
    severity_agreement = _score_severity(result_by_attr, baselin\
    e_by_attr)
    cause_alignment = _score_cause_alignment(result_by_attr, bas\
    eline_by_attr)
    mitigation_specificity = _score_mitigation_specificity(resul\
    t_by_attr)
    research_grounding = _score_research_grounding(result, resul\
    t_by_attr)

    total = completeness + severity_agreement + cause_alignment \
    + mitigation_specificity + research_grounding
    return {
        "total_score": round(total, 2),
        "subscores": {
            "completeness": round(completeness, 2),
            "severity_agreement": round(severity_agreement, 2),
            "cause_alignment": round(cause_alignment, 2),
            "mitigation_specificity": round(mitigation_specifici\
    ty, 2),
            "research_grounding": round(research_grounding, 2),
        },
        "summary": build_score_summary({
            "total_score": total,
            "subscores": {
                "completeness": completeness,
                "severity_agreement": severity_agreement,
                "cause_alignment": cause_alignment,
                "mitigation_specificity": mitigation_specificity\
    ,
                "research_grounding": research_grounding,
            },
        }),
    }
```

scripts/llm_benchmark_common.py

```
def _score_completeness(result: dict[str, Any], expected_attrs: \
    list[str]) -> float:
    """Score structural completeness (40 pts max).

    Supports two output formats:
    - Multi-attribute: result has 'reference_audit_spec' + 'qual\
    itative' list.
    - Single-attribute: result has 'qualitative' list with one i\
    tem; no audit spec.

    Both formats award points for the same semantic qualities — \
    identification,
    diagnosis depth, element coverage, justification depth, rese\
    arch backing,
    and attribute coverage — using format-appropriate field chec\
    ks.
    """
    score = 0.0
    qualitative = result.get("qualitative", [])
    spec = result.get("reference_audit_spec")

    if spec:
        # □□ multi-attribute format □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□\
    □□□□□□□□□
        if spec.get("name"):
            score += 4
        if len(spec.get("core_metrics", [])) >= 4:
            score += 8
        if len(spec.get("required_response_elements", [])) >= 4:
            score += 8
        if spec.get("why_this_spec"):
            score += 5
        if len(spec.get("supporting_research", [])) >= 2:
            score += 5
    else:
        # □□ single-attribute format □□□□□□□□□□□□□□□□□□□□□□□□□□□□□\
    □□□□□□□□□
        # Score against the first qualitative entry (there shoul\
    d be exactly one).
        item = qualitative[0] if qualitative else {}
        # 4 pts: attribute identified
        if item.get("attribute"):
            score += 4
        # 8 pts: what_is_wrong is substantive (□100 chars □ "□4 \
    metrics discussed")
        if len(str(item.get("what_is_wrong", ""))) >= 100:
            score += 8
        # 8 pts: all four narrative fields non-empty
        if all(item.get(f) for f in ("what_is_wrong", "why_is_wr\
    ong", "how_to_fix", "severity")):
            score += 8
        # 5 pts: why_is_wrong is substantive (□100 chars □ justi\
    fication paragraph)
        if len(str(item.get("why_is_wrong", ""))) >= 100:
            score += 5
        # 5 pts: □2 supporting research entries
        if len(item.get("supporting_research", [])) >= 2:
            score += 5

    # □□ attribute coverage (both formats) □□□□□□□□□□□□□□□□□□□□□□□\
    □□□□□□□
    result_attrs = {row.get("attribute") for row in qualitative}
    if set(expected_attrs).issubset(result_attrs):
        score += 10

    return min(score, 40.0)
```

scripts/llm_benchmark_common.py

```
def _score_severity(result_by_attr: dict[str, dict[str, Any]], b\
    aseline_by_attr: dict[str, dict[str, Any]]) -> float:
    if not baseline_by_attr:
        return 0.0
    matches = 0
    for attr, baseline in baseline_by_attr.items():
        got = result_by_attr.get(attr, {})
        if _normalize_severity(got.get("severity", "")) == _norm\
    alize_severity(baseline.get("severity", "")):
            matches += 1
    return 20.0 * (matches / len(baseline_by_attr))
```

scripts/llm_benchmark_common.py

```
def _score_cause_alignment(result_by_attr: dict[str, dict[str, A\
    ny]], baseline_by_attr: dict[str, dict[str, Any]]) -> float:
    if not baseline_by_attr:
        return 0.0
    matched = 0
    total = 0
    for attr, baseline in baseline_by_attr.items():
        why = (result_by_attr.get(attr, {}).get("why_is_wrong", \
    "") or "").lower()
        labels = baseline.get("root_cause_labels", [])
        for label in labels:
            total += 1
            plain = label.replace("_", " ")
            if plain in why:
                matched += 1
    if total == 0:
        return 10.0
    return 15.0 * (matched / total)
```

scripts/llm_benchmark_common.py

```
def _score_mitigation_specificity(result_by_attr: dict[str, dict\
```

```
    [str, Any]]) -> float:
    if not result_by_attr:
        return 0.0
    hits = 0
    for row in result_by_attr.values():
        text = (row.get("how_to_fix", "") or "").lower().replace\
    ("-", "").replace("_", "")
        if any(keyword.replace("-", "").replace("_", "") in text\
     for keyword in ALGORITHM_KEYWORDS):
            hits += 1
    return 15.0 * (hits / max(len(result_by_attr), 1))
```

scripts/llm_benchmark_common.py

```
def _score_research_grounding(result: dict[str, Any], result_by_\
    attr: dict[str, dict[str, Any]]) -> float:
    score = 0.0
    if len(result.get("reference_audit_spec", {}).get("supportin\
    g_research", [])) >= 2:
        score += 5
    attr_hits = 0
    for row in result_by_attr.values():
        if len(row.get("supporting_research", [])) >= 1:
            attr_hits += 1
    if result_by_attr:
        score += 10.0 * (attr_hits / len(result_by_attr))
    return score
```

F CLAIM → ARTIFACT → COMMAND

Generated by `scripts/artifact_map.py`; `tests/test_artifact_map.py` asserts every artifact exists and the table is current. Two headline numbers were previously published with no artifact behind them, and both were wrong; this table is the control that makes a third detectable.

Table 4: Every load-bearing claim, the artifact it traces to, and the command that regenerates it.

claim	artifact	regenerate with
AIF360 default DI 1.2072 vs oriented 0.9862; Fairlearn 0.8284 on the column as encoded	`artifacts/consolidated/aif360_default_repro.txt`	`python3.11 scripts/aif360_default_repro.py`
In-vivo orientation artifact: DI 0.0 (CRITICAL) vs 0.9931 on Lending Club gender	`artifacts/lending_club/fairness/fairness_summary.json`	`python3.11 run_pipeline.py german_credit hmda lending_club`
Marginal DI per attribute (Table 2), incl. banded age 0.9791 vs age-62+ 0.8897	`artifacts/hmda/fairness/fairness_summary.json`	`python3.11 run_pipeline.py german_credit hmda lending_club`
Full-population race x sex cells with Wilson CIs; 6/12 indeterminate; defensible set	`artifacts/consolidated/interval_estimates.json`	`python3.11 scripts/interval_estimates.py --emit-tex`
Reference-convention shift: 0.6734 to 0.7012, 0.8632 to 0.8988 vs White x Male	`artifacts/consolidated/interval_estimates.json`	`python3.11 scripts/interval_estimates.py --emit-tex`
Split vs population: 5/11 cells suppressed; 1.1504 (n=4) to 0.6734 (n=31); 0.1917 to 0.8815	`artifacts/consolidated/population_comparison.json`	`python3.11 scripts/compare_populations.py`
Subsampling rates (Fig. 1, Table 3, subsampling appendix): false clearance 83.8% point / 100.0% EB; floors 100% wrong worst cell; Wilson abstains 90.3% (m=2196)	`artifacts/consolidated/subsample_study.json`	`python3.11 scripts/subsample_study.py`
Population construction: 20,000 to 10,978; Race Not Available 4,140 rows (20.7%); raw-approval DI with dropped categories retained	`artifacts/consolidated/population_attrition.json`	`python3.11 scripts/hmda_attrition.py`
Refusal detector: 6 of 9 constrained-cycle responses flagged, zero refusals	`artifacts/llm_benchmark/gpt-4o`	`python3.11 scripts/report_track_q.py`
Gemini means 19.79 (all) / 47.50 (excl. zero scores) / 72.50 (content-bearing); 18 empty results; 4 empty-but-scored	`artifacts/consolidated/prompt_sensitivity_check.json`	`python3.11 scripts/recheck_prompt_sensitivity.py`
Provider block recorded on the invalid measurement	`configs/research_guardrails.json`	`cat configs/research_guardrails.json`
Fabrication flags 17 vs 4 on identical packs; 14 of 21 derivable from context	`artifacts/consolidated/RESULTS.md`	`python3.11 scripts/llm_benchmark.py --replay (see RESULTS.md S4.1)`
Track Q agreement 75.0% decomposing to 33.3/91.7/100.0; 7 over- vs 2 under-escalations	`artifacts/consolidated/track_q_analysis.json`	`python3.11 scripts/report_track_q.py`
Prompt-sensitivity spreads 14.67/9.0 vs 0.0/1.88; spread excl. constrained cycle 6.67/1.25; dip decomposition	`artifacts/consolidated/prompt_sensitivity_check.json`	`python3.11 scripts/recheck_prompt_sensitivity.py`
Track H: 27/27 citations verified against the retrieved pool	`artifacts/consolidated/track_h_evaluation.json`	`python3.11 scripts/evaluate_track_h.py`
German Credit AgeGroup x Sex cell table; under-40 female 0.8201	`artifacts/consolidated/interval_estimates.json`	`python3.11 scripts/interval_estimates.py --emit-tex`
Out-of-fold scoring: RF acc 0.8002 / AUC 0.8018 over 10,978 rows, 5-fold, seed 42	`hmda_dataset/metrics/out_of_fold_metrics.json`	`python3.11 hmda_dataset/scripts/train_models.py --full-population`